ADSP PROJECT

Synthetic Dataset Generation for Car-Wash Hazard Detection



TEAM 4

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FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems



VALUE PROPOSITION

- **WHO** car-wash operators and their ML teams
- **NEED** a hazard camera, but no real footage
- **WHAT** an automatic synthetic-data pipeline
- **BENEFIT** a detector that becomes trainable
- **HOW** edit → filter hallucinations → SAM 3 labels

OBJECTIVES

OUTPUT

- Automatic pipeline: edit, validate, label
- Labelled synthetic hazard dataset
- Quality report per image and class

OUTCOME

- Detector trainable without real footage
- ML teams know which classes to trust, from measured quality
- Little manual labelling: people review only a sample

IMPACT

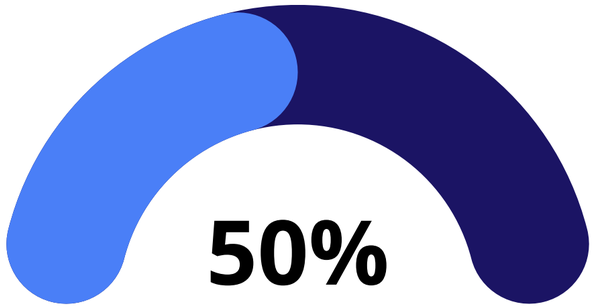
- Fewer jams and less damage at car washes
- No needless stops for drivers





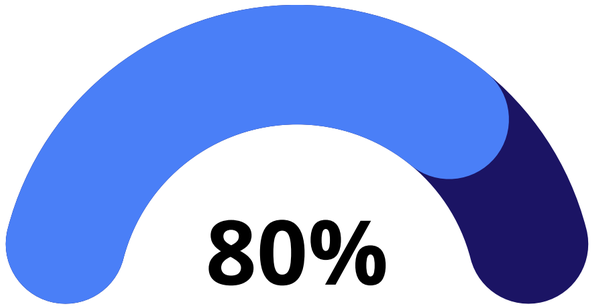
KPIs

GENERATION



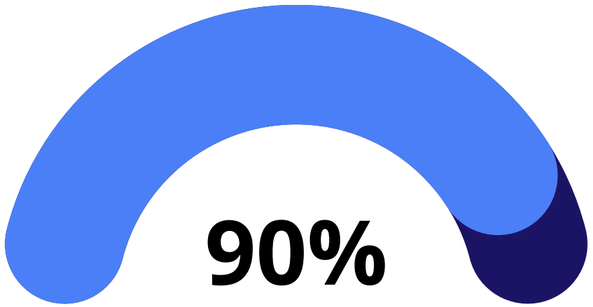
- At least 50% of each class's images pass validation
- Classes below this are dropped

VALIDATION



- Agreement between the automatic checks and human review on a sample

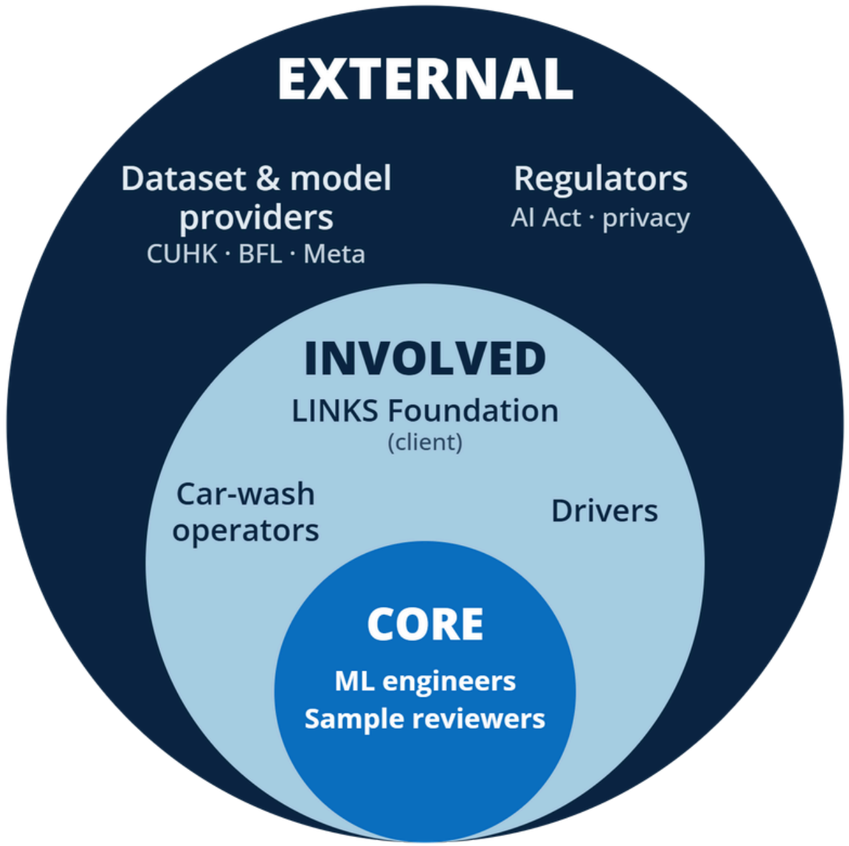
LABELLING



- At least 90% of SAM 3 masks usable
- IoU ≥ 0.5 against a hand-labelled sample

STAKEHOLDERS

From who uses the dataset to who sets the rules





PERSONAS



Mike

ML Engineer



John

Car Wash Operator

PERSONAS



MIKE *ML ENGINEER*	**TRAIN**	**EVALUATE**	**DEPLOY**
USER ACTION	**Train** a hazard detection **model** using labeled car-wash images	**Evaluate** the model's ability to **recognize** different **hazard** classes	**Integrate** the trained model into a car-wash **monitoring system**
POSSIBLE PAIN	**Lack** of diverse, labeled **images** of hazards	**Limited** real-world **hazard examples** make reliable evaluation difficult	**Poor generalization** may lead to **missed hazards** in real-world scenarios
OPPORTUNITY	**Use** our synthetic dataset containing labeled hazard scenarios	**Test** the **model** across multiple hazard types	**Improve** hazard **recognition** using model trained on **diverse** data

PERSONAS



JOHN *CAR WASH OPERATOR*	**MONITOR**	**IDENTIFY**	**PREVENT**
USER ACTION	**Watch** incoming **vehicles** at the car-wash entrance	**Spot** external **hazards** (roof boxes, bike racks, open doors, towballs)	Stop **unsafe** vehicles **before they ente**r the wash bay
POSSIBLE PAIN	**Lack** of a clear **360° view** leaves **blind spots** around the roof, rear, and far side	**Subtle hazards**, like an open fuel flap, are easy to **miss**	Unnoticed hazards **jam** or **damage** the automatic car-wash machinery
OPPORTUNITY	**Cover** every vehicle **angle** automatically using entrance **cameras**	**Highlight** any of the 23 **hazard types** on screen as the car approaches	**Trigger** instant visual **alerts** to halt the wash before damage occurs

DATASET

CompCars

The CompCars dataset contains data from two scenarios, including images from web-nature and surveillance-nature.

The images of the web-nature are collected from car forums, public websites, and search engines.

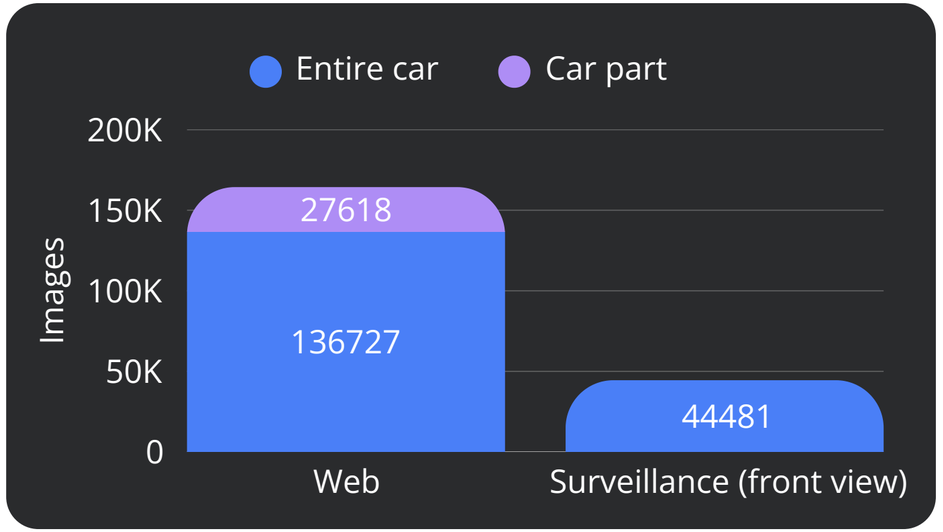
The images of the surveillance-nature are collected by surveillance cameras. The data of these two scenarios are widely used in the real-world applications.

≈208.826 images

≈20 GB

Dataset dimension

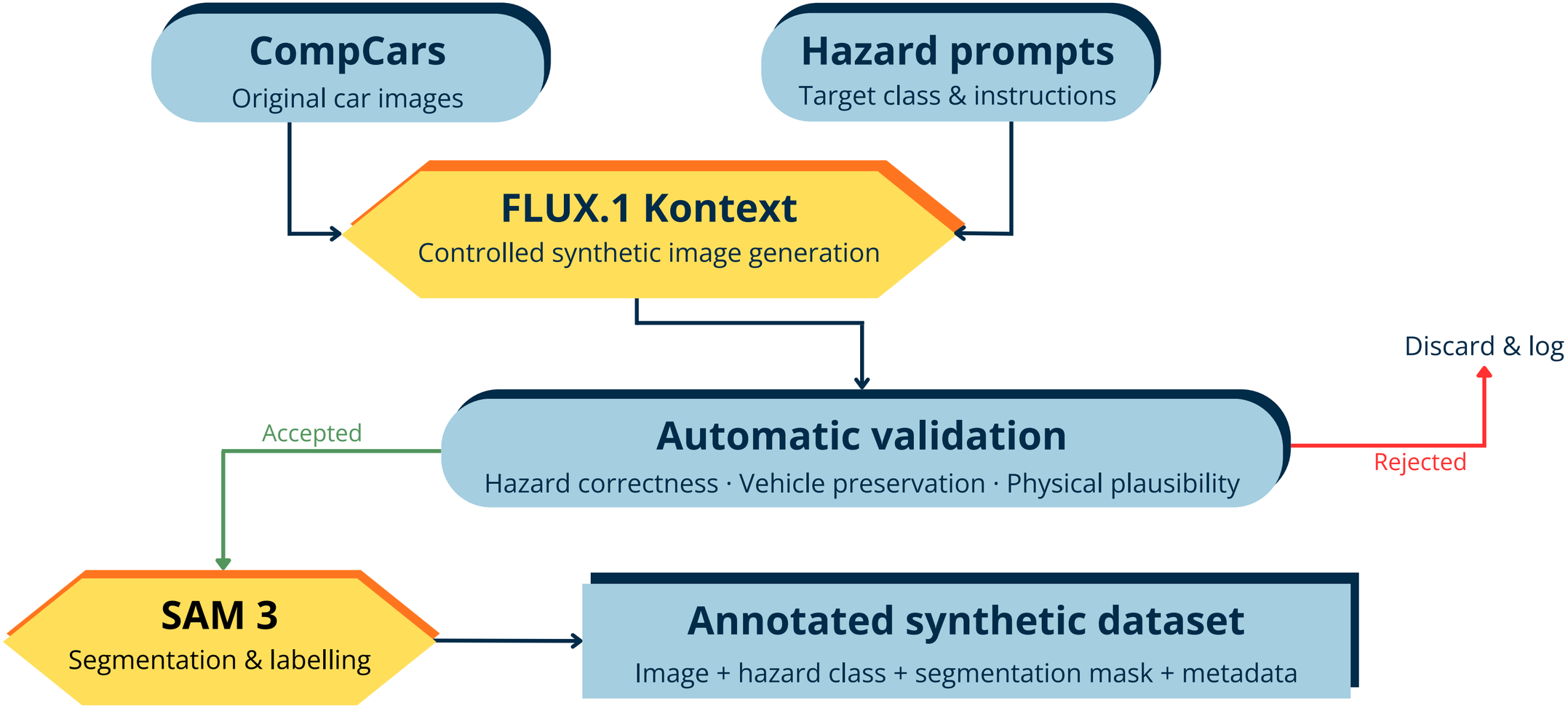




NEED	FUNCTIONAL REQUIREMENTS	ACCEPTANCE
R1 – Generate hazard scenarios	The system **generates synthetic images** representing hazardous situations (open doors, hoods, trunks, or people near vehicles)	Generated images are **reviewed** to verify that the requested hazards are **clearly represented**
R2 – Generate labeled annotations	The system produces **segmentation masks** and **labels** identifying the safety-critical objects in each synthetic image	A reviewer verifies that **masks** and **labels** correctly **match** the corresponding **objects**
R3 – Validate and export dataset	The system checks annotation consistency and **exports the images and labels**	Passes **integrity checks** and loads in the target training pipeline



HOW DOES OUR SYSTEM WORK?



HOW COULD THE OUTPUT BE USED IN PRACTICE?

Synthetic dataset
Training data

CCTV Camera
Real-time input

Trained Hazard Detector
e. g. lightweight YOLO model

No Hazard
Proceed normally

Hazard detected
Safety Alarm



PIPELINE MAIN RISKS

1. **Low Generation Success Rate**
   Image generation failures and unrealistic outputs

2. **Unreliable Automatic Validation**
   Incorrectly accepted or rejected images

3. **Inaccurate Segmentation Masks**
   Incorrect hazard boundaries



Prompt: Open the rear trunk.
Generated output: Front hood opened instead

WHAT'S NEXT?



GENERATION

- Optimize prompts
- Expand generation to all hazard class

VALIDATION

- Hallucination detection
- Human sample review

ANNOTATION

- SAM 3 optimization
- Evaluate mask quality (IoU)

THANK YOU FOR YOUR ATTENTION!